MTH636/SDS636 (2026-27, ODD SEMESTER)

PRACTICE PROBLEM SET 1

1. NASH EQUILIBRIUM

1. A Nash equilibrium s^* is *strict* if every deviation undertaken by a player yields a definite loss for that player, i.e., $u_i(s^*) > u_i(s_i, s^*_{-i}))$ for each player $i \in N$ and each strategy $s_i \in S_i \setminus \{s^*_i\}$.

   (a) Prove that if the process of iterative elimination of strictly dominated strategies results in a unique strategy vector s^*, then s^* is a strict Nash equilibrium.

   (b) Find a game that has at least one equilibrium, but in which iterative elimination of dominated strategies yields a game with no equilibria.

2. In the following three-player game, Player 1 chooses a row (A or B), Player 2 chooses a column (a or b), and Player 3 chooses a matrix (α, β, or γ). Find all the equilibria of this game.

$\boldsymbol{\alpha}$	a	b
A	$0,0,5$	$0,0,0$
B	$2,0,0$	$0,0,0$

$\boldsymbol{\beta}$	a	b
A	$1,2,3$	$0,0,0$
B	$0,0,0$	$1,2,3$

$\boldsymbol{\gamma}$	a	b
A	$0,0,0$	$0,0,0$
B	$0,5,0$	$0,0,4$

3. Find an example of a game $G = \langle N, \{S_i\}_{i\in N}, \{u_i\}_{i\in N}\rangle$ such that the game $\hat{G}$, derived from G by elimination of one strategy in one player's strategy set, has an equilibrium that is not an equilibrium in the game G.

4. Find an example of a strategic form game G and of an equilibrium s^* of that game such that for each player $i \in N$ the strategy s^*_i is dominated.

5. A two-player game is symmetric if the two players have the same strategy set $S_1 = S_2 = S$ and the payoff functions satisfy $u_1(s_1, s_2) = u_2(s_2, s_1)$ for each $s_1, s_2 \in S$. Suppose (s_1, s_2) is an equilibrium. Is (s_2, s_1) also an equilibrium?

6. Rohit (Player 1), and Virat (Player 2), are business partners. Each of the partners has to determine the amount of effort he or she will put into the business, which is denoted by e_i , $i = 1, 2$, and may be any non-negative real number. The cost of effort e_i for Player i is ce_i , where $c > 0$ is equal for both players. The success of the business depends on the amount of effort put in by the players; the business's profit is denoted by $r(e_1, e_2) = e_1^{\alpha_1} e_2^{\alpha_2}$, where $\alpha_1, \alpha_2 \in (0, 1)$ are fixed constants known by Rohit and Virat, and the profit is shared equally between the two partners. Each player's utility is given by the difference between the share of the profit received by that player and the cost of the effort he or she put into the business. Answer the following questions:

   (i) Describe this situation as a strategic-form game.

   (ii) Find all the Nash equilibria (in pure strategies) of the game.

7. Two competing coffee house chains, Pete's Coffee and Caribou Coffee, are seeking locations for new branch stores in Cambridge. The town is comprised of only one street, along which all the residents live. Each of the two chains therefore needs to choose a single point within the interval $[0, 1]$, which represents the exact location of the branch store along the road. It is assumed that each resident will go to the coffee house that is nearest to his place of residence. If the two chains choose the exact same location, they will each attract an equal number of customers. Each chain, of course, seeks to maximize its number of customers.

   To simplify the analysis required here, suppose that each point along the interval $[0, 1]$ represents a town resident, and that the fraction of residents who frequent each coffee house is the fraction of points closer to one store than to the other.

   (i) Describe this situation as a two-player strategic-form game.

   (ii) Prove that the only equilibrium in this game is that given by both chains selecting the location $x = \frac{1}{2}$.

   (iii) Prove that if three chains were to compete for a location in Cambridge, the resulting game would have no equilibrium. (Under this scenario, if two or three of the chains

choose the same location, they will split the points closest to them equally between them.)

8. Consider a game

$$G = \langle N = \{1,2\}, (S_i)_{i\in N}, (u_i)_{i\in N}\rangle$$

where $|S_i| = 2$ for $i \in \{1,2\}$.

Prove or disprove: Suppose G has exactly one pure strategy Nash equilibrium. There cannot be any other Nash equilibrium in mixed strategies.

9. Consider a finite game

$$G = \langle N, (S_i)_{i\in N}, (u_i)_{i\in N}\rangle$$

and its mixed extension Γ. Define the minmax value for Player i as

$$\overline{v}_i = \min_{\sigma_{-i}\in\Sigma_{-i}} \max_{\sigma_i\in\Sigma_i} U_i(\sigma_i, \sigma_{-i}).$$

$\overline{v}_i$ is the lowest possible payoff that the other players can force on Player i.

Prove or disprove: For any NE σ^* in mixed strategies,

$$U_i(\sigma^*) \geq \overline{v}_i.$$

2. TWO-PLAYER ZERO-SUM GAMES

1. (a) Let A be an arbitrary $m \times n$ matrix game. Show that any two saddle points of the matrix A must have the same value. In other words, if (i,j) and (k,l) are two saddle points, show that $a_{ij} = a_{kl}$.

   (b) Let A be a matrix game where both the players have 4 pure strategies. Assume that $(1,1)$ and $(4,4)$ are saddle points. Show that A has at least two other saddle points.

   (c) Give an example of a matrix game where both the players have 4 pure strategies such that there are exactly three saddle points.

2. Let A and B be two finite-dimensional matrices with positive entries. Show that the game

A	0
0	B

has no value in pure strategies. (Each 0 here represents a matrix of the proper dimensions, such that all of its entries are 0.)

3. Consider the following matrix game:

$$\begin{bmatrix} 3 & 2 & 0 \\ 1 & 2 & 2 \\ 0 & 2 & 4 \\ 0 & 3 & 1 \end{bmatrix}$$

   (i) Determine all the max-min (pure) strategies of both the players. What can you conclude about the value of the game in mixed strategies?

   (ii) The value of the game is $\frac{12}{7}$. Use this to give an argument why player 2 will put zero probability on column 2 in any maxmin strategy.

4. The value of the two-player zero-sum game given by the matrix A is 0. Is it necessarily true that the value of the two-player zero-sum game given by the matrix $-A$ is also 0? If your answer is yes, prove this. If your answer is no, provide a counterexample.

5. (**Symmetric games:**) A matrix game A is called *symmetric* if $A = -A^T$. Prove that the value of a symmetric game is zero and that the sets of max-min strategies of players 1 and 2 coincide.